

PDF EDITOR PROJECT PLAN

12-Week Development Timeline

With QA & Bi-Weekly Client Deliveries

Project Overview

- Hybrid PDF Editor (client-side UI + backend services)
- Full clone of pdfeditor.org with enhanced backend features
- Cloud document storage with cross-device access
- Backend-powered Office conversions and advanced features

Timeline Overview

Week	Deliverables	QA Gate
1-3	Core Viewer, Auth, Backend Setup	Internal QA + Client Review 1
4-6	Drawing Tools, Signatures, Forms	Internal QA + Client Review 2
7-9	Page Operations, Organization	Internal QA + Client Review 3
10-12	Conversions, Security, Polish	Final QA + Launch Ready

Detailed Weekly Breakdown

WEEKS 1-3: Core Viewer + Basic Editing + Backend Foundation

Focus: Establish foundation with frontend UI and backend infrastructure

Frontend Development:

- Landing page with navbar, hero section, feature cards, footer
- User authentication (Login/Signup) with My Account drawer
- Hamburger menu: My PDFs, Create New, Open, Save, Export
- PDF viewer with zoom controls, undo/redo, performance panel
- Basic text editing with floating toolbar
- Right sidebar with background, stroke, opacity, link, position controls

Backend Development:

- API foundation setup (Node.js/Express or similar)
- AWS S3 integration for document storage
- Database setup for user/document metadata
- Authentication endpoints (JWT/OAuth)
- Document upload/save/retrieve endpoints
- **Export as PDF endpoint**
- **IndexedDB fallback for offline viewing**

Deliverables:

- Fully functional landing page + editor UI with cloud document persistence

WEEKS 4-6: Rich Objects + Signature + Form Mode

Focus: Add interactive drawing and form creation capabilities

Frontend Features:

- Shape tools (rectangle, circle, line, polygon)
- Image upload and placement
- Free brush drawing tool with pressure sensitivity
- Eraser tool with adjustable size
- Signature creation modal (draw or upload)
- Form mode toggle with form field support
- Layers panel with full visibility/lock/delete controls
- **Contextual right sidebar styling panel**

Backend Enhancements:

- **Document versioning (save multiple edits)**
- Signature storage and retrieval

Deliverables:

- **All annotation and drawing tools working; form mode operational**

WEEKS 7-9: Page Operations + Organize

Focus: Enable comprehensive PDF page manipulation

Frontend Features:

- Split PDF modal (Range/Pages modes, customized/fixed)

- Thumbnail sidebar with drag-to-reorder pages
- Resize pages (custom dimensions)
- Rotate pages (left, right, 180°)
- **Add page numbers to PDF**
- Watermark pages
- Add background to pages
- Add blank pages
- Remove pages from document

Backend Operations:

- **PDF split processing**
- Page rotation/resize endpoints
- Watermark/background application

Deliverables:

- Complete page organization and manipulation suite

WEEKS 10-12: Optimization + Security + Conversions + Polish

Focus: Advanced features, security, and production readiness

Frontend Features:

- Compress dialog (High/Balanced/Light + Custom quality/resize)
- Password protect/unprotect with 128/256-bit encryption
- Redact tool (permanent content removal)
- Flatten PDF (form-field locking)
- Full Export menu: PDF, JPG, PNG, Word, Excel, PowerPoint
- Extract images from PDF (as ZIP)
- Find & Replace across full document
- **Pricing page with 7-day trial (\$0.70) → \$19.99/month**
- Responsive layout + mobile touch support

Backend Features:

- High-fidelity conversions: PDF ↔ DOCX, XLSX, PPTX
- PDF compression with quality preservation

- Standard PDF encryption (AES 256-bit, ARC4 128-bit)
- Billing integration (Stripe) with recurring subscription
- Privacy and security messaging
- Final performance optimization and bug fixes

Deliverables:

- Production-ready, fully-featured PDF editor with all premium features

QA & Testing Schedule

Weekly QA Gate (Friday of each week)

Every Friday:

- Internal QA testing of completed features
- Automated test execution
- Cross-browser compatibility checks
- Performance benchmarking
- Security vulnerability scanning
- Bug fixes and refinements before client review

Bi-Weekly Client Deliveries

Delivery 1 (End of Week 3):

- Live staging environment with Weeks 1-2 deliverables
- Landing page, authentication, PDF viewer, basic editing
- Functional test account + demo video walkthrough

Delivery 2 (End of Week 6):

- Updated staging with all drawing, signature, and form features
- Layers panel fully operational
- QA report with test results and known issues

Delivery 3 (End of Week 9):

- Complete page organization and manipulation suite
- Split, resize, rotate, watermark, add page numbers
- Performance metrics and browser compatibility report

Final Delivery (End of Week 12):

- Production-ready application with all features
- Stripe billing integration live with trial subscription
- Full documentation and deployment guide
- Security audit report
- Launch-ready with monitoring and support plan

Team & Resources

Role	Responsibility	Allocation
Frontend Lead	UI/UX implementation, React components, state management	100% - Full-time
Backend Lead	API design, AWS integration, database, conversions	100% - Full-time
QA Engineer	Testing, bug tracking, regression analysis	80% - Part-time
Project Manager	Timeline tracking, stakeholder updates, blockers	100% - Full-time

Key Milestones & Decision Points

- **Week 3 Checkpoint:** Confirm backend infrastructure stability + frontend responsiveness
- **Week 6 Checkpoint:** Validate annotation toolset + form capabilities
- **Week 9 Checkpoint:** Confirm page operations performance on large files
- **Week 12 Launch:** Production deployment with monitoring and support

Known Risks & Mitigation

Risk	Impact	Mitigation
Large file handling	Performance degradation >100MB	Chunked uploads, progressive rendering
Office conversions	Quality/fidelity issues	Third-party library evaluation + fallback handlers

Browser compatibility	Edge cases in older browsers	Automated cross-browser testing weekly
-----------------------	------------------------------	--

Explicitly Out of Scope

- Repair functionality
- Dedicated Merge PDF feature (thumbnail drag-to-reorder only)
- AI features
- Real-time collaboration
- Version history / rollback

End of Project Plan